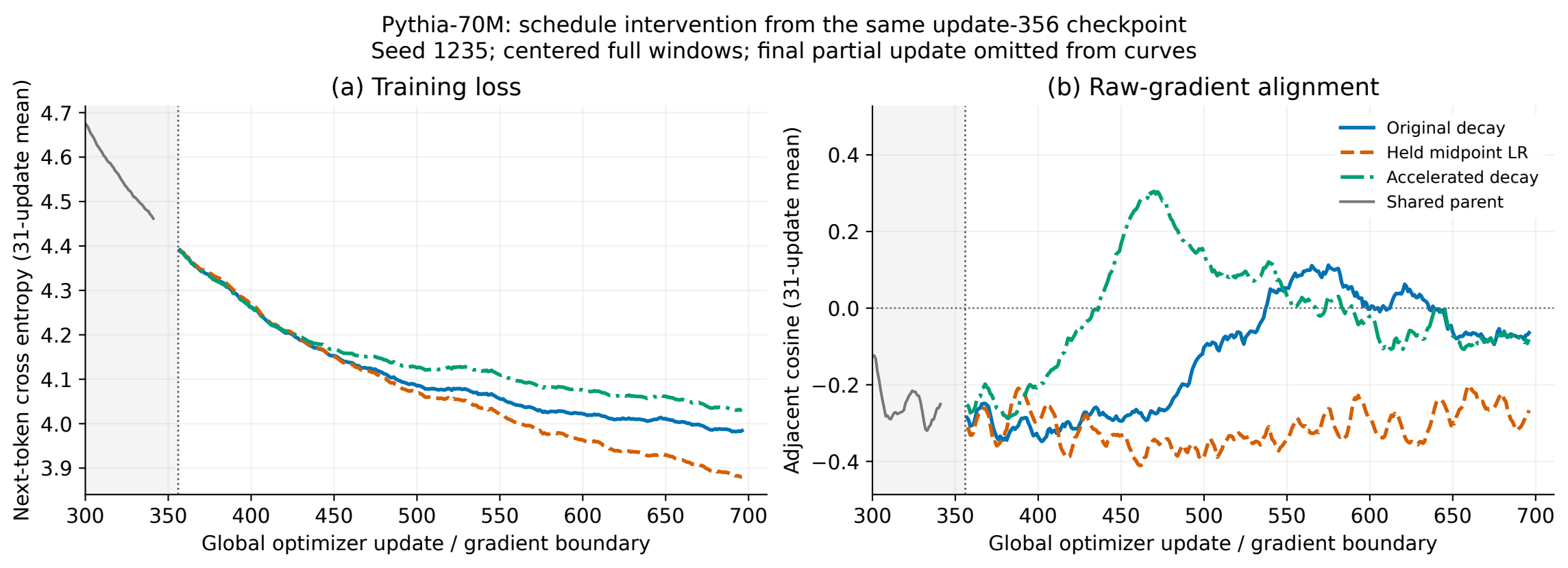

Pythia-70M: schedule intervention from the same update-356 checkpoint
Seed 1235; centered full windows; final partial update omitted from curves
(a) Training loss
Next-token cross entropy (31-update mean)
4.7
4.6
4.5
4.4
4.3
4.2
4.1
4.0
3.9
300
350
400
450
500
550
600
650
700
Global optimizer update / gradient boundary
(b) Raw-gradient alignment
Adjacent cosine (31-update mean)
0.4
0.2
0.0
−0.2
−0.4
Original decay
Held midpoint LR
Accelerated decay
Shared parent
Global optimizer update / gradient boundary